

DBMS: ASSIGNMENT 8

MongoDB- Aggregation and Indexing

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Batch:A4

Case Study: Employee Appraisal System

```
test> use employee1
switched to db employee1
employee1> db.employee.insertMany([
| { emp_id: 1, name: "ABC", salary: 10000, rating: 5 },
| { emp_id: 2, name: "XYZ", salary: 15000, rating: 4 },
| { emp_id: 3, name: "John", salary: 20000, rating: 3 },
| { emp_id: 4, name: "Riya", salary: 18000, rating: 2 }
|])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('G9dde7Gf4ccf9f1bcc3G82d1'),
    '1': ObjectId('G9dde7Gf4ccf9f1bcc3G82d2'),
    '2': ObjectId('G9dde7Gf4ccf9f1bcc3G82d3'),
    '3': ObjectId('G9dde7Gf4ccf9f1bcc3G82d4')
  }
}
employee1> db.employee.find()
[
  {
    _id: ObjectId('G9ddeGda2c0182125G3G82d1'),
    emp_id: 1,
    name: 'ABC',
    salary: 10000,
    rating: 5
```

```
},
{
  _id: ObjectId('G9ddeGda2c0182125G3G82d2'),
  emp_id: 2,
  name: 'XYZ',
  salary: 15000,
  rating: 4
},
{
  _id: ObjectId('G9ddeGda2c0182125G3G82d3'),
  emp_id: 3,
  name: 'John',
  salary: 20000,
  rating: 3
},
{
  _id: ObjectId('G9ddeGda2c0182125G3G82d4'),
  emp_id: 4,
  name: 'Riya',
  salary: 18000,
  rating: 2
},
{
  _id: ObjectId('G9dde7Gf4ccf9f1bcc3G82d1'),
  emp_id: 1,
  name: 'ABC',
  salary: 10000,
  rating: 5
},
{
  _id: ObjectId('G9dde7Gf4ccf9f1bcc3G82d2'),
  emp_id: 2,
  name: 'XYZ',
  salary: 15000,
  rating: 4
}
```

```

    },
    {
      _id: ObjectId('G9dde7Gf4ccf9f1bcc3G82d3'),
      emp_id: 3,
      name: 'John',
      salary: 20000,
      rating: 3
    },
    {
      _id: ObjectId('G9dde7Gf4ccf9f1bcc3G82d4'),
      emp_id: 4,
      name: 'Riya',
      salary: 18000,
      rating: 2
    }
  ]
employee1> db.employee.aggregate([
| { $group: { _id: null, total_salary: { $sum: "$salary" } } }
| ])
[ { _id: null, total_salary: 120000 } ]
employee1>
db.employee.aggregate([
| { $group: { _id: null, avg_salary: { $avg: "$salary" } } }
| ])
[ { _id: null, avg_salary: 15750 } ]
employee1>
db.employee.aggregate([
| { $group: { _id: null, max_salary: { $max: "$salary" } } }
| ])
[ { _id: null, max_salary: 20000 } ]
employee1>
db.employee.aggregate([
| { $group: { _id: "$rating", total_salary: { $sum: "$salary" } } }
| ])
[
  { _id: 2, total_salary: 30000 },
  { _id: 3, total_salary: 40000 },
  { _id: 5, total_salary: 20000 },

```

```
{ _id: 4, total_salary: 30000 }  
]  
employee1> db.employee.createIndex({ salary: 1  
}) salary_1  
employee1> db.employee.createIndex({ rating:  
-1 }) rating_-1  
employee1> db.employee.getIndexes()  
[  
  { v: 2, key: { _id: 1 }, name: '_id_' },  
  { v: 2, key: { salary: 1 }, name: 'salary_1' },  
  { v: 2, key: { rating: -1 }, name: 'rating_-1' }  
]  
employee1>
```